Fiscal Year:	FY 2009	**Task Last Updated:**	FY 03/23/2012
PI Name:	Albert, Mitchell S Ph.D.		
Project Title:	Hyperpolarized Noble Gas Magnetic Resonance Imaging		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Physiology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:	Note: PI moved to U Mass Medical School in 2007; previously at Brigham and Women's Hospital/Harvard University (9/2008)		
Project Type:	Ground	**Solicitation / Funding Source:**	01-OBPR-07
Start Date:	09/18/2007	**End Date:**	09/17/2009
No. of Post Docs:		**No. of PhD Degrees:**	1
No. of PhD Candidates:	2	**No. of Master' Degrees:**	1
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Received no-cost extension to 9/17/2009 (from 9/17/2008) per J. Dardano/JSC (9/2008)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Mulkern , Robert (Children's Hospital Boston) Reilly , John Joseph (Brigham and Women's Hospital)		
Grant/Contract No.:	NNX07AP82G		
Performance Goal No.:			
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Task Description:	A novel biomedical imaging technique, hyperpolarized (HP) noble gas magnetic resonance imaging (MRI) is capable of generating high-resolution images of the lung, heart, and brain by using the spin-1/2 noble gases, 3He and 129Xe. The HP gases, created via spin-exchange optical pumping, have greatly enhanced MR detectability. Unlike conventional proton MRI, HP gas MRI does not require a large magnetic field and, therefore, can be performed in very low magnetic fields (~ 0.01 Tesla). Thus, a small, portable, low-power MRI system, practical for space-flight, may be designed to implement very low field HP gas MRI. The imaging capacity of HP gas MRI itself is also relevant to space flight research. Under prior NASA funding, we have obtained (i) cerebral blood flow measurements and (ii) lung gas transfer measurements using HP 129Xe MRI in rats. Additionally, in humans we have (i) obtained high-resolution images of the distal airways in lungs up to the 7th generation, (ii) created models of the three-dimensional lung airway tree, and (iii) obtained very low field (0.015 Tesla) images of the lungs, using HP 3He MRI. We now propose to apply this novel technique to investigations of the physiological and functional states of the lungs, heart, and brain. Studies will focus on regional ventilation and perfusion inhomogeneities of the lungs, patent foramen ovale and pulmonary embolism of the heart, and cerebral function and blood flow of the brain. Our proposed ground-based investigations with HP gas MRI will impact space flight research in three important ways: (i) by facilitating the acquisition of novel information about the physiological structure and function of organs and systems most affected by changes in microgravity; (ii) by developing a new technology that may have clinical applications for the diagnosis of pathologies endemic to microgravity environments; and (iii) by developing treatment strategies that address the continuum of a space-induced illness.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Today, more than ever, the understanding of pulmonary function has become critical to the treatment of disease and to the health of the community at large. Currently in the U.S., pulmonary disease affects over 20 million people, results in 123,000 deaths per year, and is the fourth leading cause of death. The economic cost of pulmonary disease is estimated to be over ten billion dollars a year. The personal cost, although not quantifiable, is no doubt enormous. The understanding of environmentally induced pulmonary illness has also taken on more importance with recent advances in farming and industry. Endotoxin exposure appears to be a growing cause of airway disease in workers in livestock and agricultural operations, while environmental exposure to asbestos is the major risk factor for development of mesothelioma (cancer of the lung pleura). And most recently, the threat of biological terrorism using exposure to inhaled toxins poses the potential for catastrophic morbidity and mortality. Given this vast landscape of costly threats to the pulmonary health of the U.S. population, the economic necessity of more accurately understanding, diagnosing, and preventing naturally occurring and environmentally induced pulmonary diseases cannot be understated. HP Noble Gas MRI is quickly moving to the forefront of clinical techniques that allow physicians to more quickly and accurately diagnose natural and environmentally induced pulmonary diseases. Rapid diagnosis and staging of disease with HP Noble Gas MRI will allow earlier therapeutic intervention and thus better health outcomes, all at a cost savings for individuals and the healthcare community. HP Noble Gas MRI will also offer cost savings by increasing the versatility, resolving power and even portability of existing imaging facilities, and thereby allow the more widespread application of MRI in major medical centers and its use in smaller clinics. Because the SNR of HP Noble Gas MRI is virtually independent of magnetic field strength, the technique does not require high magnetic fields, and can be performed with cheaper magnets. This allows HP Noble Gas MRI to be carried out in most any clinical setting that has an MRI machine. Such units could be used for low-cost, portable imaging and disease diagnosis in a variety of primary care facilities, especially those which specialize in the treatment of underserved persons, and which are adequately connected to social service infrastructure, handicapped accessible, public transportation, and other services designed to assist persons with disabling conditions (e.g., visiting nurse, home health aides, child and elder care). A portable imaging device would further benefit the very persons who, by virtue of their disabling condition, and/or inability to access quality healthcare, cannot demand the best or most optimal treatment. HP Gas MRI is also expected to have an impact of the quality and kind of research that can be carried out to study the etiology and treatment of pulmonary disease. HP 129Xe can be easily and non-invasively delivered to animals so that the same animal can be used over the course of a study. Not only does this alleviate the need for sacrificing a large number of animals, it more importantly allows the study of a disease process or experimental therapeutic treatment over time, which makes it directly relevant to the study of diseases and therapeutics in human clinical trails. We are convinced that HP Noble Gas technology will one day transform the way disease is detected, diagnosed, and treated, and we are specifically committed to developing assistive imaging technologies for a wide range of pulmonary and neurological disorders—both congenital and acquired. It is our firm hope and expectation that HP Noble Gas imaging will not only shed light on heretofore elusive disease mechanisms; it will exert a positive impact on patient outcomes, quality of life, and overall longevity.
	[Ed. note 3/23/2012: compiled from Final Report submitted to JSC Human Research Program in July 2011 and received by Task Book February 2012] Hyperpolarized (HP) noble gas MRI using 3He or 129Xe provides highly detailed images of the lungs and brain, and offers the immense advantage of providing information on physiological function, including lung ventilation (3He or 129Xe), lung perfusion (129Xe), and cerebral blood flow (129Xe). For example, HP gas MRI allows investigators to pinpoint precise sections of the lungs where gas is reaching, and where gas is not reaching (ventilation defects), something other measurements of lung function cannot do. Because of its greater sensitivity, HP gas MRI can detect pulmonary or cerebral dysfunction at earlier stages, allowing treatment or counteracting measures to be provided earlier and with greater effectiveness. In addition, HP gas MRI provides imaging of physiological function without exposing subjects to ionizing radiation or chemical contrast agents. This increases safety, and also allows investigators and/or care providers to non-invasively image subjects repeatedly over time. HP gas MRI has countless applications to evaluating vascular and pulmonary problems that could arise in human space flight. Such applications include investigating the effects of microgravity and pressure changes on the physiological functioning of the lungs, and studying the effects of microgravity and pressure changes associated with space flight on vascular functioning. In addition, HP gas MRI is ideal for evaluating aerosol drug administration, a delivery method of particular interest for space flight because it avoids the need for injections of medicines in space. In this study, we further developed hyperpolarized noble gas MRI for use in examining pulmonary and brain function in animal models and in humans, building upon previous work by the PI funded by NASA. HP gas MRI has immense potential for applications to the study of the physiology of the lungs and brain, and for helping diagnose and inform

Task Progress:	treatment for adverse reactions to microgravity and low pressure environments. Low-magnetic-field MRI, which is made possible by HP gas MRI, is particularly advantageous in this regard because it provides extraordinarily detailed images of lung and brain function with lightweight, low-cost, easy-to-operate equipment that could be used in space environments. With the assistance of generous funding from NASA, our group was able to make significant advances in many areas of HP gas MRI, including pulmonary ventilation analysis and low-magnetic-field MRI. In this study, our specific research accomplishments included the following: Ventilation Heterogeneity of Asthma Evaluated Using Hyperpolarized 3He MRI Functional Imaging of the Brain Using Hyperpolarized 129Xe MRI Using Hyperpolarized 129Xe MRI to Image Stroke Hyperpolarized 3He MRI Evaluation of Radiation Therapy for Non-small Cell Lung Cancer (NSCLC) Assessment of Ventilation Function and Airway Structure in Mesothelioma Using Hyperpolarized 3He MRI Resolving the Conflict of Hyperpolarized 129Xe T1 Values in the Rat Brain Very Low field Lung MRI using Hyperpolarized 3He Developing small, distal airway imaging to resolutions below 1mm
Bibliography Type:	Description: (Last Updated: 03/23/2012)
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